Parallel circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate branch currents to a shared potential difference.

SUCCESS CRITERIA - I can...

1. Identify branches connected across the same two nodes.
2. Use equal voltage across parallel branches.
3. Add branch currents to find supply current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Electrons in a wire drift at 0.3 mm/s while a lamp responds quickly after a switch closes. Explain why response time cannot be found by timing one electron across the wire.

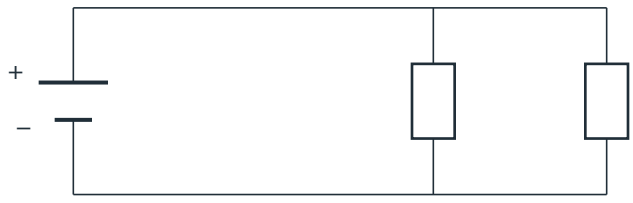
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $1/R_{total} = 1/R_1 + 1/R_2$; $I_{total} = I_1 + I_2$; $V_1 = V_2$. Parallel branches connect to the same two nodes and share voltage.

Worked reasoning: Two parallel branches of 6 and 3 ohms each have 6 V across them. Branch currents are 1 and 2 A; source current is 3 A.

Your turn: Draw the two branches and mark where the currents split and recombine.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Mark the two shared nodes in the parallel diagram. If one branch opens, can the other remain powered? Explain.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A 12 V supply powers 6 Ω and 12 Ω resistors in parallel. Label the shared nodes. Calculate branch currents, source current and equivalent resistance; check that the branch currents add to the source current.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

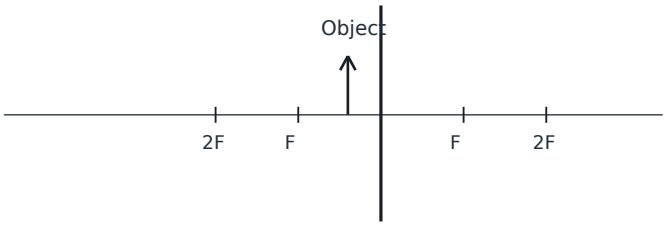
05 Explain your reasoning

Why can the equivalent resistance be smaller than either branch resistance?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A converging lens has f=20 cm and d_o=8 cm. Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$. Find d_i, magnification and image kind.
Draw the rays used in your explanation.



Converging thin lens; add principal rays.

Marzano check - circle: 1 2 3

07 Exit check

Two equal 8 Ω resistors are in parallel. Find equivalent resistance. Compare with the same resistors in series. Label the circuit and show the current path or paths.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 12 V supply feeds parallel 4 and 6 ohm resistors. Find equivalent resistance, total current and total power by two methods. Label the circuit and show the current path or paths.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3